A Spiking Neural Network and Drift-Diffusion Model Framework for P300-Based BCI

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Abstract

This document presents a conceptual architecture for integrating electroencephalography (EEG)-based P300 brain–computer interfaces (BCIs) with spiking neural networks (SNNs) and the drift-diffusion model (DDM). The proposed framework processes multi-channel EEG signals through P300 preprocessing and temporal spike encoding, followed by an adaptive leaky integrate-and-fire (Adaptive-LIF) spiking neural network. The resulting spike trajectories are used to derive an evidence signal for a DDM or accumulator model. This framework allows conventional classification performance to be evaluated alongside decision-making parameters such as drift rate, decision boundary, and non-decision time.

Keywords: Brain–Computer Interface, EEG, P300, Spiking Neural Network, Adaptive-LIF, Drift-Diffusion Model, Neuromorphic Computing, Computational Neuroscience

1 Introduction

Brain–computer interfaces (BCIs) provide a direct communication pathway between neural activity and computational systems. Among different EEG paradigms, the P300 event-related potential is widely used for studying stimulus detection and target classification.

Spiking neural networks (SNNs) provide a biologically inspired computational framework in which information is represented using discrete spike events. In addition to providing classification outputs, an SNN produces temporal spike activity that can potentially be interpreted as an evolving evidence signal.

The drift-diffusion model (DDM) provides a mathematical framework for modeling two-alternative decision making. By connecting the temporal output of an SNN to a DDM, it becomes possible

to study not only whether a target was correctly classified, but also *how evidence accumulated over time to produce a decision.*

2 Research Gap

P300-based EEG has been extensively used for target detection and Classification in brain–computer interfaces (BCIs). However, conventional BCI approaches generally evaluate the system primarily through classification-oriented measures such as accuracy, precision, and the F1-score. These measures show whether a decision is correct, but provide limited information about the underlying temporal The process by which evidence is developed and a decision is arrived at.

Drift-diffusion models (DDMs) provide a complementary framework by describing decision making as the accumulation of noisy evidence until A decision boundary is reached; previous studies have shown that EEG features can be related to DDM parameters at the single-trial level, showing that neural activity can explain variability in evidence accumulation and non-decision processes See [3]. Moreover, studies using EEG on decision making have linked neural signals such as the P300 and centroparietal positivity to computational quantities associated with evidence accumulation, the thresholds for decision and urgency [4].

In parallel, spiking neural network (SNN) models have been related to A computational and mechanistic look at DDMs. Umakantha et al. looked into how the parameters of a spiking neural model map onto drift rate, response threshold, and non-decision time of a diffusion model [5]. More recently, Jaffard et al. created a biologically plausible SNN decision model and established a theoretical relationship between spiking processes and the DDM [6].

Yet there is an important gap in these lines of research. Existing EEG–DDM studies primarily use EEG features as regressors or neural correlates of parameters estimated from behavioral data, while existing SNN–DDM studies have largely focused on simulated or abstract decision-making networks rather than P300-based EEG-driven There is still a need for a framework that deals with BCI signals. preserves the temporal spike dynamics generated from real EEG and uses these dynamics as an explicit evidence representation for decision modeling.

The proposed work addresses this gap by treating the temporal trajectory of an Adaptive-LIF SNN as an intermediate neural The relationship between P300 EEG and computational decision making. Rather than using the SNN solely as a classifier, the proposed framework retains its spike trajectory, transforms it into an evidence signal, and investigates whether this signal can account in terms of decision-making dynamics via the DDM parameters.

The central distinction is therefore not the use of a DDM or an SNN independently, but the investigation of the following computational chain:

$$\text{P300 EEG} \rightarrow \text{Temporal Spike Dynamics} \rightarrow \text{Evidence Trajectory} \rightarrow \text{DDM Parameters}. \quad (1)$$

This formulation enables the study of not only whether the BCI makes a correct decision, but also how rapidly and confidently evidence accumulates toward that decision.

3 Central Research Question

The central research question of this work is:

> *Can temporal spike dynamics extracted from P300 EEG using an Adaptive-LIF spiking neural network serve as a biologically interpretable evidence signal for drift-diffusion modeling, and can the resulting decision parameters explain trial-level variability in BCI decision accuracy and decision time?*

This question can be decomposed into three experimentally testable sub-questions:

1. **Neural evidence representation:** Can the temporal spike trajectory of the Adaptive-LIF SNN provide an evidence signal that discriminates target and non-target P300 responses?
2. **Decision dynamics:** Do DDM parameters such as drift rate (v), boundary separation (a), and non-decision time (t_0) systematically vary across correct and incorrect trials, or across fast and slow decisions?
3. **Temporal efficiency:** How early in the EEG epoch does the accumulated SNN-derived evidence become sufficient to reach a reliable decision boundary?

4 Research Objectives

The primary objectives of the proposed framework are:

1. Develop an Adaptive-LIF SNN for processing temporally encoded P300 EEG signals.
2. Preserve and characterize the temporal spike trajectory produced by the SNN rather than relying exclusively on the final classification output.
3. Construct an evidence representation from the temporal spike activity and use it as input to a DDM or accumulator model.
4. Estimate and analyze the DDM parameters

$$v, \qquad a, \qquad t_0$$

   in relation to P300-based target detection.
5. Investigate whether DDM parameters explain trial-level differences in classification correctness and decision time.

6. Investigate whether progressive accumulation of SNN-derived evidence can enable an early decision before the complete EEG epoch is processed.

7. Examine the relationship between SNN temporal dynamics and computational decision parameters to determine whether neuromorphic representations can provide an interpretable bridge between neural signals and decision behavior.

5 Proposed Architecture

The proposed architecture consists of four major processing stages:

1. EEG acquisition and P300 preprocessing,

2. temporal spike encoding,

3. Adaptive-LIF spiking neural network processing,

4. DDM-based evidence accumulation and decision modeling.

The overall architecture is illustrated in Figure **??**.

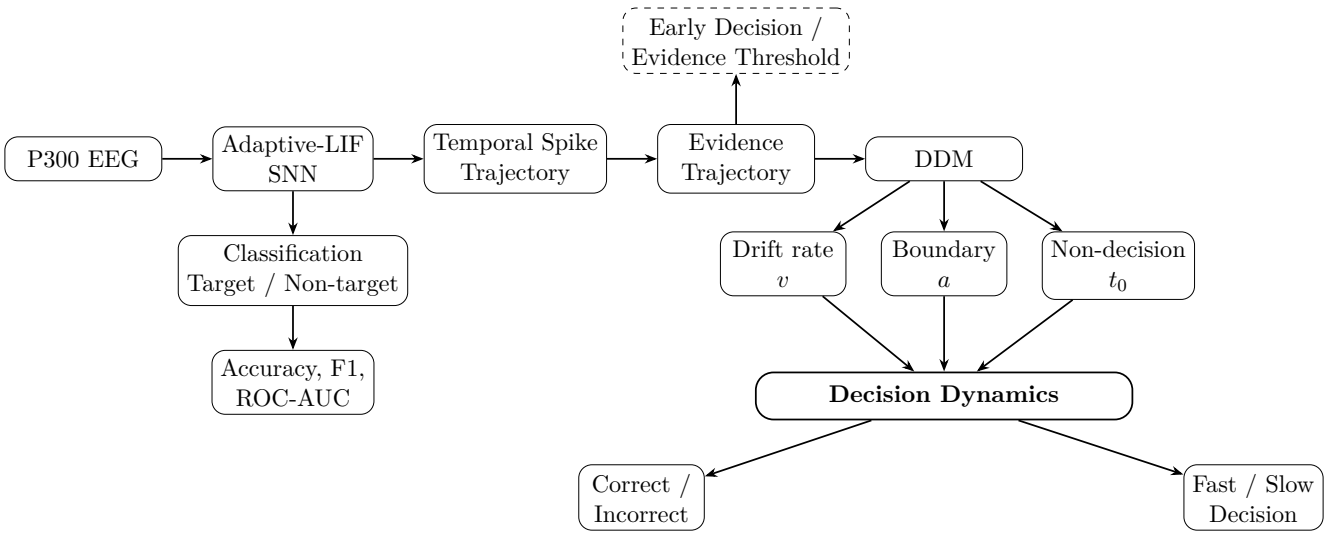


Figure 1: Conceptual framework for interpreting P300 EEG through an Adaptive-LIF spiking neural network and drift-diffusion model. Temporal spike activity is transformed into an evidence trajectory, which is used to estimate decision parameters and characterize trial-level decision dynamics.

6 EEG and P300 Processing

The first stage of the proposed framework receives an 8-channel EEG signal. The raw EEG data are preprocessed to remove artifacts and obtain the relevant P300 response.

The preprocessing stage may include filtering, artifact removal, epoch extraction, baseline correction, and normalization. The resulting P300-related signal is then provided to the temporal encoding stage.

7 Temporal Spike Encoding

To convert the continuous EEG representation into spike-based activity, a temporal encoding mechanism is applied. In the proposed architecture, Time-to-First-Spike (TTFS) encoding is considered.

In TTFS encoding, the timing of the first spike represents the magnitude of the input signal. A stronger input can therefore produce an earlier spike, while a weaker input produces a later spike.

The encoded spike trains are subsequently provided to the Adaptive-LIF spiking neural network.

8 Adaptive-LIF Spiking Neural Network

The Adaptive-LIF SNN processes the temporally encoded EEG signals. Unlike a conventional artificial neural network, the SNN explicitly represents temporal neural activity through spikes.

The SNN produces two important types of information.

8.1 Classification Output

The first output is the conventional classification result, such as:

- Target stimulus
- Non-target stimulus

Classification performance can be evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and related metrics.

8.2 Spike Trajectory

The second output is the temporal spike trajectory generated by the SNN. Instead of using only the final classification label, the evolution of spike activity over time is retained.

This temporal information can subsequently be transformed into an evidence signal for decision modeling.

9 Drift-Diffusion Model

The spike trajectory is transformed into an evidence signal that is supplied to a drift-diffusion model (DDM) or related accumulator model.

A simplified DDM can be represented as

$$dx(t) = v\,dt + s\,dW(t), \tag{2}$$

where $x(t)$ represents accumulated evidence, v is the drift rate, s represents the noise scale, and $W(t)$ denotes a Wiener process.

The accumulated evidence evolves until it reaches a decision boundary.

9.1 DDM Parameters

The proposed architecture aims to estimate three principal decision parameters:

- **Drift rate (v):** represents the average rate at which evidence accumulates toward a decision.
- **Boundary separation (a):** represents the amount of evidence required before making a decision.
- **Non-decision time (t_0):** represents the time associated with processes outside the decision accumulation itself, such as sensory encoding and motor response.

10 Expected Outputs

The proposed framework provides two complementary levels of analysis.

10.1 Machine Learning Performance

The SNN can be evaluated using conventional classification metrics, including:

$$\text{Accuracy}, \quad \text{Precision}, \quad \text{Recall}, \quad F_1, \quad \text{ROC-AUC}. \tag{3}$$

10.2 Decision-Making Parameters

The temporal spike activity can additionally be used to estimate:

$$v, \qquad a, \qquad t_0. \tag{4}$$

This provides a bridge between neuromorphic computation and computational models of decision making.

11 Conclusion

This framework combines EEG-based P300 analysis, temporal spike encoding, adaptive spiking neural networks, and drift-diffusion modeling into a unified architecture.

The key advantage of this approach is that the SNN is not treated only as a classifier. Its temporal spike activity can also be interpreted as a source of evolving evidence, which can be connected to a DDM to study decision formation.

This provides a potential computational bridge between neuromorphic representations of neural activity and behavioral models of decision making.

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